PVsyst V8.0.21

PVsyst - Simulation report

Standalone system

Project: GROUP 5 EE-367

Variant: New simulation variant

Standalone system with batteries

System power: 11.34 kWp

OBUASI CAMPUS - Ghana

GROUP 5

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Author

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Project summary

Geographical Site
OBUASI CAMPUS
Ghana

Situation

Latitude	6.20 °(N)
Longitude	-1.67 °(W)
Altitude	252 m
Time zone	UTC

Project settings

Albedo	0.20

Weather data
OBUASI CAMPUS
Meteonorm 8.2, Sat=100% - Synthetic

System summary

Standalone system

Orientation #1
Fixed plane
Tilt/Azimuth 10 / 0 °

Standalone system with batteries

User's needs
Daily household consumers
Constant over the year
Average 41 kWh/Day

System information

PV Array

Nb. of modules	42 units
Pnom total	11.34 kWp

Battery pack

Technology	Lithium-ion, LFP
Nb. of units	10 units
Voltage	48 V
Capacity	1000 Ah

Results summary

Useful energy from solar	14804 kWh/year	Specific production	1305 kWh/kWp/year	Perf. Ratio PR	66.53 %
Missing Energy	285 kWh/year	Available solar energy	17783 kWh/year	Solar Fraction SF	98.11 %
Excess (unused)	2537 kWh/year				

Table of contents

Project and results summary 2
General parameters, PV Array Characteristics, System losses 3
Detailed User's needs 4
Main results 5
Loss diagram 6
Predef. graphs 7

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General parameters

Standalone system | **Standalone system with batteries**

Orientation #1
Fixed plane
Tilt/Azimuth 10 / 0 °

Models used
Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

User's needs
Daily household consumers
Constant over the year
Average 41 kWh/Day

PV Array Characteristics

PV module
Manufacturer Generic
Model AURA F270 V2.0
(Original PVsyst database)
Unit Nom. Power 270 Wp
Number of PV modules 42 units
Nominal (STC) 11.34 kWp
Modules 21 string x 2 In series
At operating cond. (50°C)
Pmpp 10.21 kWp
U mpp 56 V
I mpp 181 A

Total PV power
Nominal (STC) 11 kWp
Total 42 modules
Module area 68.2 m²
Cell area 61.9 m²

Controller
Universal controller
Technology MPPT converter
Temp coeff. -5.0 mV/°C/Elem.
Converter
Maxi and EURO efficiencies 97.0 / 95.0 %
Battery Management control
Threshold commands as SOC calculation
Charging SOC = 0.96 / 0.80
Discharging SOC = 0.10 / 0.35

Battery Storage

Battery
Manufacturer Generic
Model ZXDC48 FB100B3
Technology Lithium-ion, LFP
Nb. of units 10 in parallel
Discharging min. SOC 10.0 %
Stored energy 44.8 kWh
Battery Pack Characteristics
Voltage 48 V
Nominal Capacity 1000 Ah (C10)
Temperature Given monthly values

Array losses

Thermal Loss factor
Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses
Global array res. 5.2 mΩ
Loss Fraction 1.50 % at STC

Serie Diode Loss
Voltage drop 0.7 V
Loss Fraction 1.1 % at STC

Module Quality Loss
Loss Fraction -0.75 %

Module mismatch losses
Loss Fraction 0.50 % at MPP

Strings Mismatch loss
Loss Fraction 0.15 %

IAM loss factor
Incidence effect (IAM): Fresnel smooth glass, n = 1.526

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.998	0.981	0.948	0.862	0.776	0.636	0.402	0.000

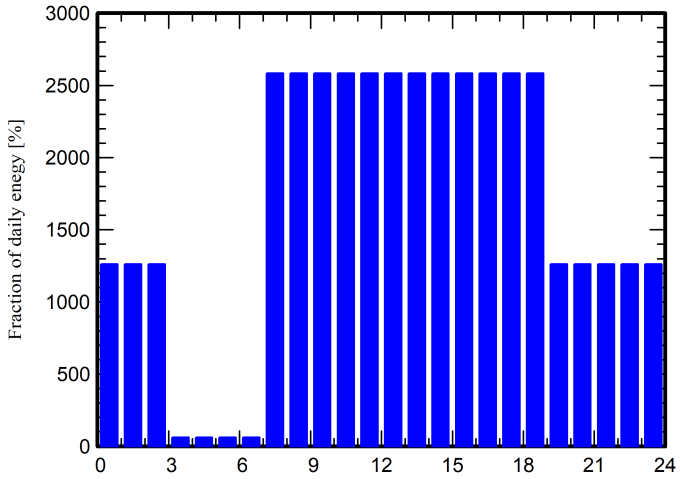
Detailed User's needs

Daily household consumers, Constant over the year, average = 41 kWh/day

Annual values

	Nb.	Power	Use	Energy
		W	Hour/day	Wh/day
Lamps LED	12	12/lamp	12.0	1728
Laptops	20	60/app	20.0	24000
Ceiling fans	7	75/app	12.0	6300
Routers	6	10 tot	24.0	1440
Projectors	2	327 tot	12.0	7848
Stand-by consumers			24.0	24
Total daily energy				41340

Hourly distribution



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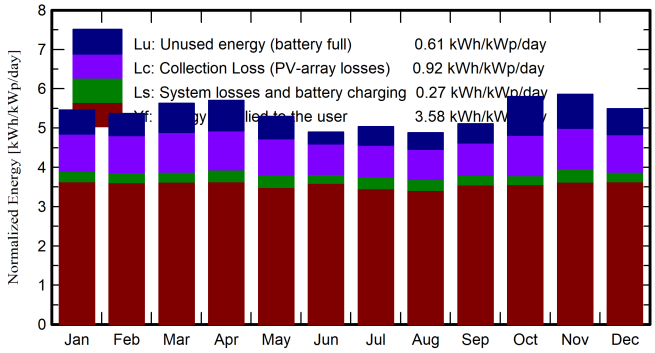
Main results

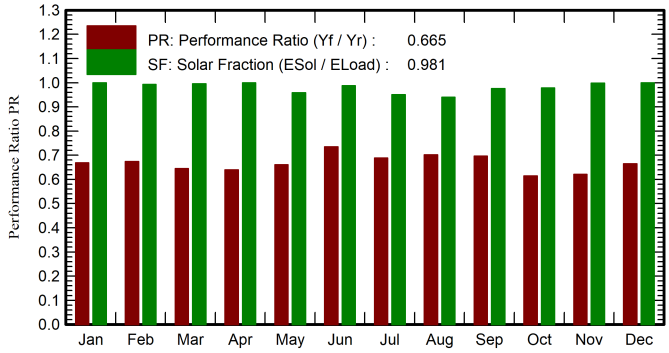
System Production

Useful energy from solar	14804 kWh/year	Perf. Ratio PR	66.53 %
Available solar energy	17783 kWh/year	Solar Fraction SF	98.11 %
Excess (unused)	2537 kWh/year		
Loss of Load		**Battery ageing (State of Wear)**	
Time Fraction	2.6 %	Cycles SOW	94.4 %
Missing Energy	285 kWh/year	Static SOW	93.3 %
		Battery lifetime	15.0 years

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor kWh/m²	GlobEff kWh/m²	E_Avail kWh	EUnused kWh	E_Miss kWh	E_User kWh	E_Load kWh	SolFrac ratio
January	158.7	164.7	1532	212	0.0	1282	1282	1.000
February	145.4	146.6	1354	176	7.9	1150	1158	0.993
March	173.1	170.1	1567	257	5.4	1276	1282	0.996
April	174.8	166.4	1543	260	0.0	1240	1240	1.000
May	173.1	158.6	1476	196	53.0	1229	1282	0.959
June	156.6	141.5	1346	99	15.6	1225	1240	0.987
July	165.0	150.8	1432	164	64.0	1218	1282	0.950
August	156.1	146.4	1387	146	77.7	1204	1282	0.939
September	153.9	148.5	1399	162	29.8	1210	1240	0.976
October	173.8	175.1	1621	345	28.3	1253	1282	0.978
November	164.6	171.3	1585	290	3.0	1237	1240	0.998
December	158.3	166.1	1541	229	0.0	1282	1282	1.000
Year	1953.2	1906.2	17783	2537	284.8	14804	15089	0.981

Legends

GlobHor	Global horizontal irradiation	E_User	Energy supplied to the user
GlobEff	Effective Global, corr. for IAM and shadings	E_Load	Energy need of the user (Load)
E_Avail	Available Solar Energy	SolFrac	Solar fraction (EUsed / ELoad)
EUnused	Unused energy (battery full)		
E_Miss	Missing energy		

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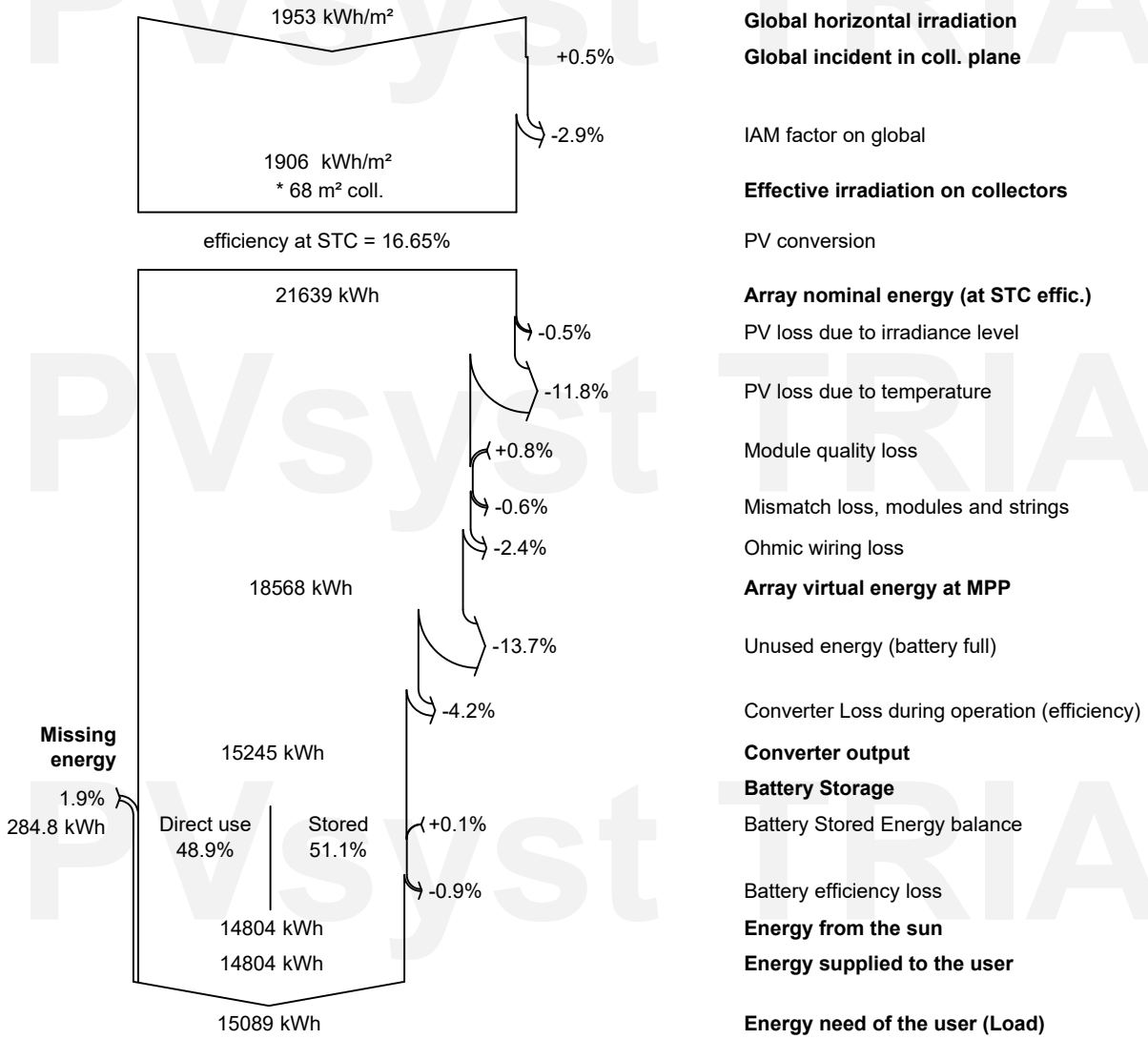
Loss diagram



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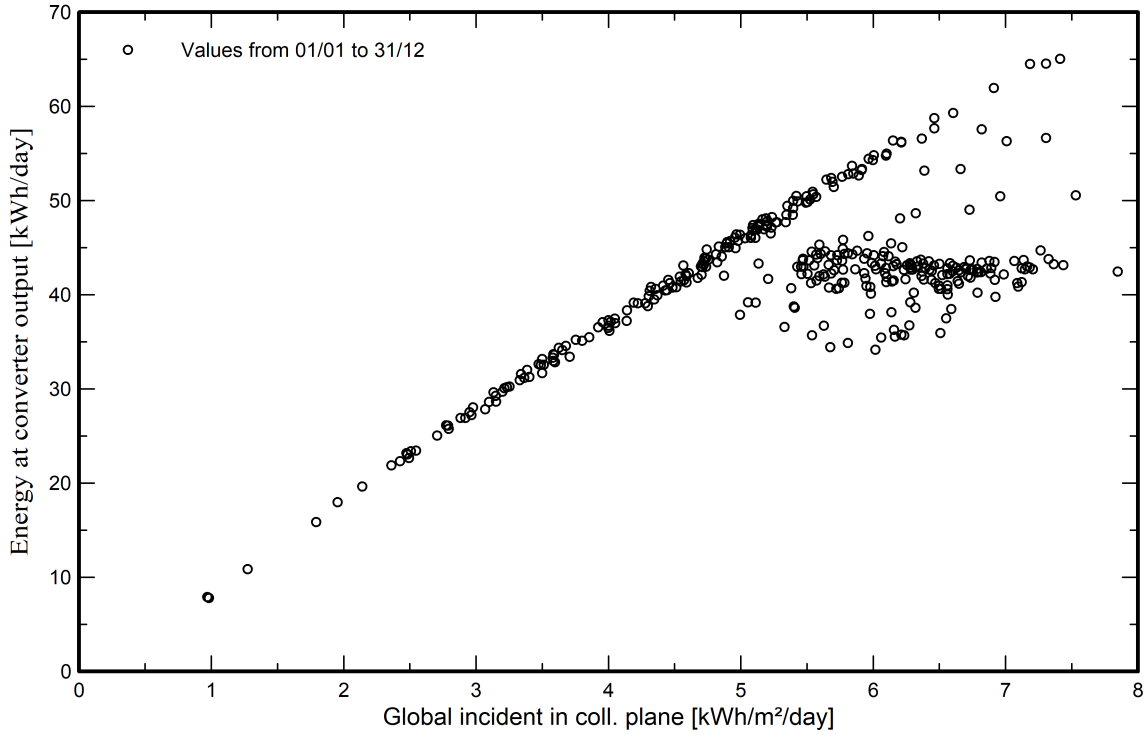
Predef. graphs

Daily Input/Output diagram



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